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Start : January 2025 **Doctoral School** : ED129 (Sciences de l'Environnement de l'île de France)

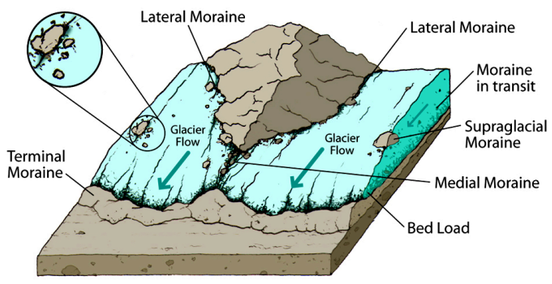
Reconstruction of the Glacier Fronts Trajectories by Stochastic Simulations

Context

1856 - Today

Mer de Glace (France)

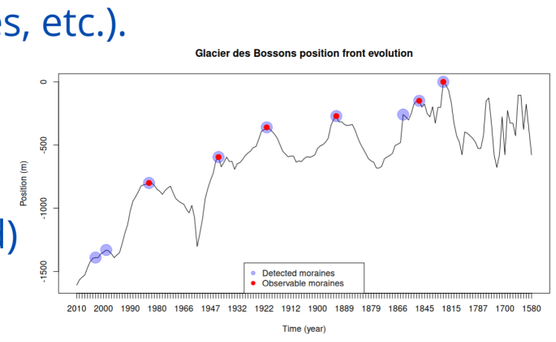
Moraine formation schema



The retreat of glaciers modifies **climatic conditions** at altitude and creates **instability** (ice falls, avalanches, etc.).

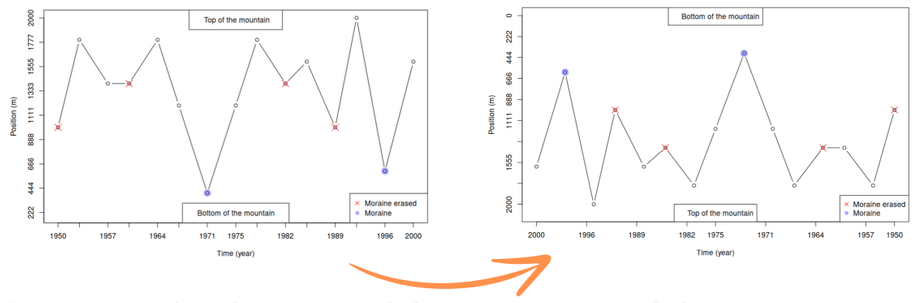
Observed data

- Recent data (less than 200 years)
- 25 glaciers (France and Switzerland)
- Annual front variation (trajectory)



Visual representation

How to link moraines and records ?



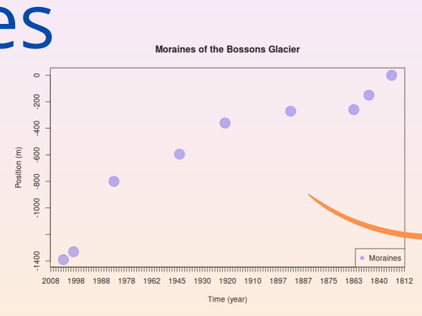
Reversing the direction of the time axis and the position axis

Definition

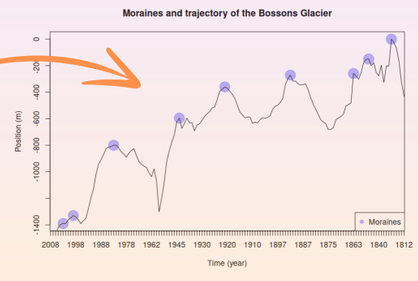
Visible **moraines** are evidence of past glacier positions.
They correspond to the **records** of the series of the **local maxima** of the front relative positions when the time axis is reversed.

Objective : Build a probabilistic model tracing glacier trajectories using the dates and positions of their moraines

Challenges : Respect the physical constraints of the moraines
Goal : Apply it to existing moraine series and retrace past trajectories to be able to identify the glaciers that are most sensitive to climate, based on long-term trends
Approach : Explore and compare 2 methods

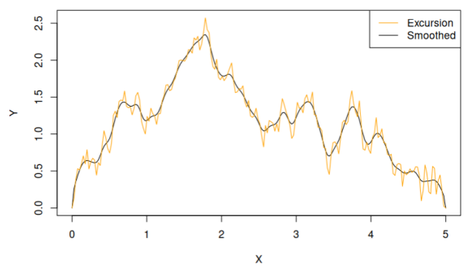


How do you go from that to this?



1- Brownian Motions

Random walk satisfying Markov property
Brownian bridge : brownian motion which is equal to zero at both ends
Brownian excursion : Nonnegative brownian bridge



Example of a brownian excursion

Model : Generate brownian bridges **between each pair of moraines**
Verwaat transform : obtain **brownian excursions and change the start and end point** while keeping brownian properties

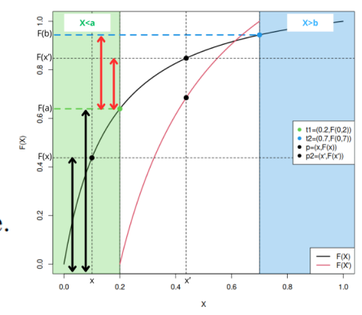
2- Folding and Gaussian simulation

Definition : Folding Principle
X a random variable and F the distribution function of X

$$X^{(F)}(a,b) = \begin{cases} F^{-1}\left(\frac{F(X)}{F(a)}(F(b)-F(a))+F(a)\right) & \text{if } X<a \\ F^{-1}\left(\frac{\bar{F}(X)}{\bar{F}(b)}(F(a)-F(b))+F(b)\right) & \text{if } X>b \\ X & \text{otherwise.} \end{cases}$$

Proposition
For $a < x < b$,

$$P(X^{(F)}(a,b) > x) = P(X > x | a < X < b)$$

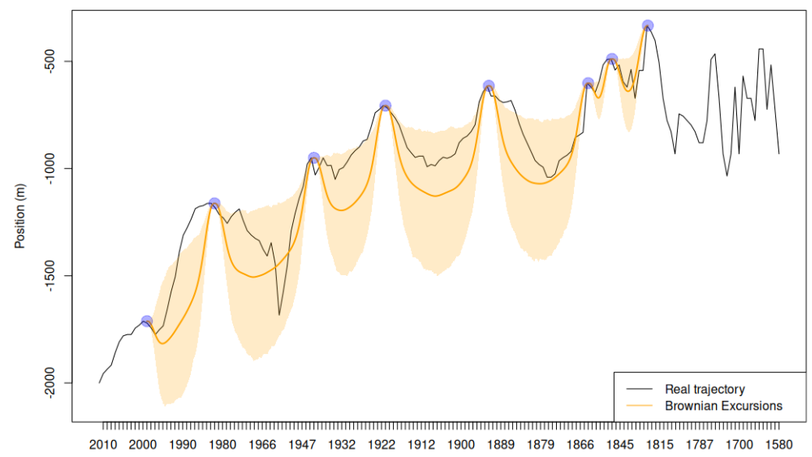


Folding Principle illustrated

Model : Use conditional simulation to create the glacier trajectory **point by point**
For each simulated point :

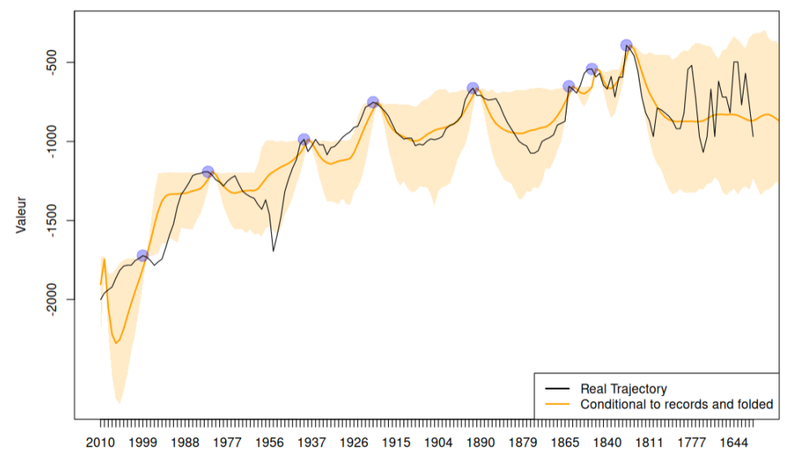
- Fold it to make sure it is **smaller than the following moraine**
- Add it to the subset used to simulate the next point

First results



Brownian Motions

Mean and 5-95% enveloppes for both methods computed on 1000 simulations



Folding and Gaussian simulation

Next steps

Questions :

- Which physical constraints should be considered ? (Volume, slope, climate data ?)
- How to compare simualted trajectories ? (score)

How to Improve of the models ? :

- Long-term front trajectories behaviour independently of climate forcing data
- Specificities of trajectories as a function of local characteristics (slope,glacier size, ...) or regional context (Alps, Himalayas, ...)?

Uses of the model :

- Inference to evaluate the parameters
- Evaluation and validation on available complete trajectories data
- Application to existing moraine series
- Interpretation in an interdisciplinary manner

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PEPR Risques (IRiMa)